SANDHIYA S

Embedded Systems Engineer

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PROFESSIONAL SUMMARY

Dedicated Electronics and Communication Engineering student with hands-on experience in microcontroller programming, IoT systems, sensor interfacing, and real-time embedded applications. Skilled in C, C++, and hardware-software integration with Arduino and ESP32 platforms. Passionate about building practical embedded and IoT solutions for industrial and consumer applications. Keen to contribute to innovative projects and grow as an embedded systems professional.

SKILLS

Programming Languages : C, C++, Java
Communication Protocols : I2C, UART, SPI
MCU Peripherals : ADC, Timer, PWM
Microcontrollers : Arduino, ESP32
Tools : Arduino IDE, VS Code, KiCad
Web Development – HTML, CSS (Basics)

INTERNSHIP EXPERIENCES

Elamani Tech Pvt. Ltd. Mar 2026 - Jun 2026
Embedded Systems Intern | R&D Department

- Worked on **Bagasse Cleaning and Overflow Prevention using Automation**, implementing sensor-based flow detection and motor control for industrial juice processing. Gained exposure to industrial automation concepts, component selection, and real-time process control system design.
- Analyzed the embedded system architecture of a commercial delivery robot.

Tamirabot Advanced Engineering Pvt. Ltd.
Trainee – (Embedded Systems) Jun 2026 - Jul 2026

- Developed a **15A Smart Wall Socket with 4G, NFC, and Temperature Monitoring**, integrating sensors and communication modules for real-time intelligent plug control and IoT-based automation for EV vehicles.

Salem Steel Plant – SAIL | Intern - Centralized Electrical Maintenance Dec 2025

- Observed centralized electrical maintenance procedures, industrial power distribution, and safety practices while observing real-time operation of heavy electrical machinery in a large-scale steel manufacturing plant.

PROJECTS

Smart Luggage Tracking System(IoT)

- *Tech Stack:* ESP32, GPS Module, IoT Sensors, Firebase
- Description: Developed an IoT- based luggage tracking system for Real-time location monitoring and travel security. Integrated GPS and sensors to detect tampering and unauthorized movement, send live updates to users, and trigger alert notifications — demonstrating practical embedded systems design and IoT integration.

AI-Based Machine Fault Prediction and Alert System (STM32)

- *Tech Stack:* STM32, NanoEdge AI Studio, Temperature & Vibration Sensors, IoT Monitoring
- Description: Designed a predictive maintenance system using STM32-based data acquisition and anomaly detection in real-time temperature and vibration data. Enabled IoT-based alerts for proactive maintenance, reducing unplanned machine downtime.

EDUCATION

B.E. Electronics and Communication Engineering 2023 - Apr 2027
Government College of Engineering, Salem | CGPA: 8.26

Higher Secondary 2022 - 2023
Jayarani Girl's Higher Secondary School, Salem | Percentage: 94.1%

AWARDS & CERTIFICATIONS

- Attended workshop on Hands-on Training on Electronics Measuring Instruments.
- Won 3rd place in Paper Presentation at a national level technical symposium.